Zecpath AI Full System Simulation Report

Objective

Validate the complete AI hiring pipeline from resume upload to final hiring decision and compare AI-generated decisions with human evaluation.

Simulation Scope

The following pipeline was tested:

1. Resume Upload
2. ATS Parsing & Scoring
3. AI Screening
4. HR Interview AI
5. Technical Interview AI
6. Machine Test Evaluation
7. Behavioral Analysis
8. Integrity Detection
9. Cross-Round Aggregation
10. Final Recommendation AI
11. Hiring Report Generation

Test Dataset

Parameter	Value
Total Candidates	50
Roles Covered	Backend, Frontend, DevOps, Data Science
Experience Levels	Fresher to Senior
Total Evaluations	~400 AI Decisions

Sample Output


```
{
  "candidate_id": "C20001",
  "scores": {
    "ats": 78,
    "screening": 74,
```

```
      "hr": 80,
      "technical": 85,
      "machine_test": 82
    },
    "behavior": {
      "risk_level": "Low Risk"
    },
    "integrity": {
      "risk_level": "Moderate Risk"
    },
    "final_score": 80.5,
    "decision": "Selected"
  }
```


AI vs Human Evaluation

Metric	Result
Decision Match Rate	88%
Score Correlation	0.86
False Positives	7%
False Negatives	5%

Stage Accuracy

Stage	Accuracy
ATS	90%
Screening	85%
HR Interview	87%
Technical Interview	89%
Machine Test	92%

Conclusion

The simulation demonstrates that the Zecpath AI recruitment pipeline performs consistently across all recruitment stages with an overall decision accuracy of **88%** and is suitable for controlled production deployment.